Sachin Ashok

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LinkedIn Google Scholar GitHub

Research Statement

I am a researcher in networked systems, focusing on improving the **performance and observability of distributed applications and cloud infrastructure**. My work combines systems design, large-scale experimentation, and algorithmic techniques to build practical systems for distributed request tracing, performance diagnosis, and slack-aware scheduling, often **without requiring application modifications**. More recently, I have been exploring scheduling and resource management challenges in AI infrastructure and agentic systems. My research has led to publications at **SIGCOMM, SIGMETRICS, and HotNets**.

Education

2021 – 2027 Ph.D. in Computer Science, GPA: 4.0 — Expected 05/27
University of Illinois Urbana-Champaign
Advisors: **Prof. Radhika Mittal, Prof. Philip Brighten Godfrey**

2014 – 2018 B.Tech. in Computer Science and Engineering
National Institute of Technology, Trichy

Publications

[1] **The Cloud Should Adapt to Its Applications, Not the Other Way Around!**
Sachin Ashok*, Karuna Grewal*, Gangmuk Lim, Talha Waheed, Gerard Matthew, Aditya Prerepa, Brighten Godfrey, Justin Hsu, Radhika Mittal
In Proceedings of ACM HotNets, 2026.

[2] **TraceWeaver: Distributed Request Tracing for Microservices Without Application Modification**
Sachin Ashok, Vipul Harsh, Brighten Godfrey, Radhika Mittal, Srinivasan Parthasarthy, Larisa Shwartz
In Proceedings of ACM SIGCOMM, 2024.

[3] **Murphy: Performance Diagnosis of Distributed Cloud Applications**
Vipul Harsh, Wenxuan Zhou, Sachin Ashok, Radhika N. Mysore, Brighten Godfrey, Sujata Banerjee
In Proceedings of ACM SIGCOMM, 2023.

[4] **Data-Driven Network Path Simulation with iBox**
Sachin Ashok, Shubham Tiwari, Nagarajan Natarajan, Venkat Padmanabhan, Sundararajan Sellamanickam
In Proceedings of ACM SIGMETRICS, 2022.

[5] **Fast and Efficient Look-Ups via Data-Driven FIB Designs**
Sachin Ashok*, Aditi Partap*, Ammar Tahir* (* = equal contribution)
In Proceedings of ACM SIGCOMM FIRA Workshop, 2022.

[6] **Leveraging Service Meshes as a New Network Layer**
Sachin Ashok, P. Brighten Godfrey, Radhika Mittal
In Proceedings of ACM HotNets, 2021.

[7] **iBox: Internet in a Box**
Sachin Ashok, Sai Surya Duvvuri, Nagarajan Natarajan, Venkat Padmanabhan, Sundararajan Sellamanickam, Johannes Gehrke
In Proceedings of ACM HotNets, 2020.

[8] **RL for Bandwidth Estimation and Congestion Control in Real-Time Communications**
Joyce Fang, Martin Ellis, Bin Li, Siyao Liu, Yasaman Hosseinkashi, Michael Revow, Albert Sadovnikov, Ziyuan Liu, Peng Cheng, Sachin Ashok, David Zhao, Ross Cutler, Yan Lu, Johannes Gehrke
In Proceedings of NeurIPS MLForSystems Workshop, 2019.

Under Submission/ In Preparation

[1] **Enabling Slack Based Scheduling for Microservice Systems**
Sachin Ashok, Brighten Godfrey, Radhika Mittal
Under Submission.

[2] **Multiparty Load Balancing**
Talha Waheed, Sachin Ashok, Brighten Godfrey, Radhika Mittal, R. Srikanth
Under Submission.

[3] **TierX: Adaptive Prioritization to Meet SLO Guarantees for Diverse Workloads**
Gerard Matthew, Sachin Ashok, Brighten Godfrey, Radhika Mittal
In Preparation.

Research and Industry Experience

2024 **Microsoft Research, Redmond** — **Research Intern**
Mentor: **Prateesh Goyal**, **Kevin Hsieh**.
Explored adaptive scheduling and admission control techniques to improve efficiency of telemetry pipelines.

2018 – 2021 **Microsoft Research, India** — **Research Fellow**
Mentors: **Venkat Padmanabhan**, **Nagarajan Natarajan**, **Sundar Sellamanickam**, **Johannes Gehrke**.
Designed a data-informed network simulator for recreating the behavior of paths in a target network.

2017 **Samsung R&D, India** — **Intern**
Developed an end-to-end object localization framework using deep learning models like CNNs.

2016 **Mozilla Organization** — **Intern**
Implemented Blake2 (a cryptographic hash function) support for Mozilla's NSS library.

Teaching, Mentoring, and Service

Teaching

- **Teaching Assistant**, CS 435: Cloud Networking, University of Illinois Urbana-Champaign, Spring 2025
  - Received the **Outstanding Teaching Assistant Award.**
  - Designed and ran a load balancing contest with a public leaderboard and prizes to help students build intuition for practical load balancing strategies and tradeoffs.
- **Guest Lecturer**, CS 425: Distributed Systems, University of Illinois Urbana-Champaign, Spring 2024
- **Guest Lecturer**, ECE 598: Modern Cloud Infrastructure, University of Illinois Urbana-Champaign, Spring 2025

Mentoring

- **Research Mentor**, University of Illinois Urbana-Champaign, 2021 – Present
  - Mentoring 2 PhD students and 2 undergraduates on research in networked systems and cloud infrastructure.
- **Research Mentor**, Microsoft Research India, 2019 – 2024
  - Mentored 4 research fellows; mentees co-authored papers at SIGMETRICS and HotNets, and joined PhD programs at University of Washington and University of Texas at Austin.
  - Guest Lecture on Applying to Graduate School, 2022
- **Academic Mentor**, Delta Force (Programming Club), NIT Trichy, 2015 – 2018
  - Mentored 6 junior students on competitive programming and software development.

Academic Service

- **External Reviewer**, ACM SIGCOMM 2025
- **Artifact Evaluation Committee**, ACM SIGCOMM 2025, 2026
- **Reviewer**, IEEE Network Magazine 2024
- **CS Graduate Ambassador**, University of Illinois Urbana-Champaign
- **Invited Panelist**, NetworkingChannel 2024

Invited Talks

Distributed Tracing without the Pain!, **KubeCon 2022**, Detroit, USA

iBox: Internet in a Box for Realistic Network Simulation, **Microsoft Bandwidth Control Workshop 2020**, Redmond

Awards and Honors

Outstanding Teaching Assistant Award, CS 435 (Spring 2025), UIUC CS Department

Student Travel Grants, SIGMETRICS 2022, SIGCOMM 2023, HotNets 2023

Project Demo Invite, Microsoft TechFest, Seattle, 2020

Finalist, Pragyan CTF, NIT Trichy (Top 1% of participants), 2018

Selected Participant, IISc CSA Summer School (Top 100 chosen nationwide), 2017

Selected Participant, IMSc Summer Research Program (Top 25 chosen nationwide), 2017

Selected Participant, Mozilla Winter of Security (Among 12 teams chosen globally), 2016

Skills and Background

Languages: C++, C, Python, Bash (*advanced*); Golang, JavaScript (*proficient*)

Tools: Kubernetes, Docker, eBPF, Istio, Protobuf, ns-2/3, gRPC, TensorFlow, NetEm, Wireshark

Courses: Advanced Networking (CS 538), Advanced OS (CS 523), Reliability of Cloud Systems (CS 598XU), High-Speed & Programmable Networks (CS 598RM), Advanced Distributed Systems (CS 525)